What Counts as Progress?

Activity indicators and later assessment performance

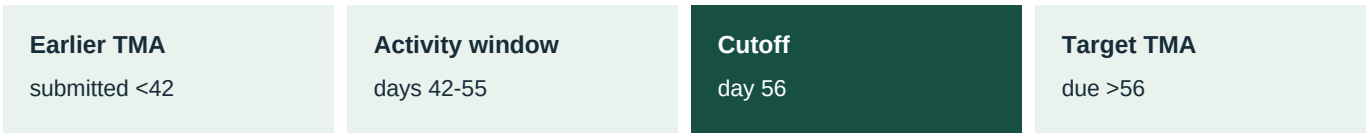
How much information do recorded platform activity counts add about later assessed performance beyond earlier performance and course context?

ELIGIBLE ENROLLMENTS	OBSERVED LATER MARKS	RESERVED EVALUATION
20,441 across 19 offerings	17,019 mark records	3,399 enrollments / 3,197 learners

The main result

Adding click volume reduced mean absolute prediction error from **11.62 to 11.30 mark points**. The reduction was **0.319 points**, with a 95% learner-cluster bootstrap interval [0.207, 0.426]. This is a modest improvement in a retrospective prediction task.

The study uses the public Open University Learning Analytics Dataset (OULAD). Activity is observed on days 42-55; the target is a later tutor-marked assessment. A locally dated analysis plan, learner-level evaluation split, missing-outcome diagnostics, and six sensitivity comparisons make the measurement choices inspectable. [1,7]



Timing rules shown schematically; widths are not durations.

Scope: later marks are assessed performance, not measured learning gain or transfer. These data contain no dashboard treatment. The results do not establish that clicks cause learning or that displaying a metric changes study choices.

Companion artifact: a fictional dashboard and practice-scoring demonstration. No participants have been recruited; no participant experiment or learning benefit is claimed. The full editable methods report, code and aggregate results accompany this PDF.

Data and cohort

Source and measurement

OULAD records anonymized registrations, assessments and daily resource interactions from 2013-2014. The official UCI archive contains 32,593 registrations and 10,655,280 activity rows. Acquisition checksums, table counts, schema checks and join checks are preserved. The release is CC BY 4.0. [1,7]

The analysis unit is an enrollment in a module offering. A learner may have several enrollments; all of that learner's records stay in one partition. No raw learner records or identifiers are included in the portfolio package.

Eligibility and assessment selection

Registration must precede day 56, with no earlier recorded withdrawal. The prior assessment is the latest scheduled TMA due before day 42; it needs a nonbanked observed mark in 0-100 and actual submission before day 42. The target is the first positive-weight TMA due after day 56. Deadline ties use the smallest assessment identifier.

Known banked target credit and target work already submitted before the cutoff are excluded. Enrollments without a target submission or mark remain missing-outcome cases; they are not assigned zero scores. Mark models require an observed target score from a submission on or after day 56.

Preparation stage	Enrollment records
All source registrations	32,593
Supported assessment schedules	30,059
Registered and not withdrawn before cutoff	24,109
Valid prior score before window	20,506
Eligible after target exclusions	20,441
Target submission recorded	17,048
Target mark observed	17,019

Coverage

The eligible cohort contains 19,028 distinct learners in six modules and 19 offerings. There are 3,422 missing target marks (16.74%): 3,393 without a submission record and 29 with a submission but no mark. The mark analysis therefore describes a selected population with prior assessed work and an observed later score.

The data provide submission and deadline dates, but no grade-release timestamp. Earlier-mark adjustment is retrospective: a live dashboard could not necessarily have displayed that mark at the chosen cutoff.

Analysis and verification

Two visible indicators

Volume sums all official interaction counts on days 42-55 and enters the model as natural log(1 + clicks). **Active days** counts days with positive recorded activity, from 0 to 14. Zero recorded activity establishes no observed platform interaction; it does not establish no study.

Model	Predictors
Reference	Prior score + course-offering fixed effects
Volume / primary	Reference + log(1 + total clicks)
Consistency / secondary	Reference + active days
Combined / secondary	Reference + both activity indicators

Reserved learners and uncertainty

A deterministic SHA-256 partition reserves approximately 20% of learners. Development includes 16,342 eligible enrollments and 13,620 observed marks; evaluation includes 4,099 eligible enrollments and 3,399 observed marks. Learner overlap is zero. Models are fitted to development marks and saved before evaluation outcome files are read. Future submission and withdrawal fields never enter prediction formulas.

The primary criterion is reference MAE minus volume MAE, in assessment-mark points. Positive values mean reduced error. Predictions are ordinary unbounded linear-model predictions; none of the four primary models produced evaluation predictions outside 0-100.

Intervals use 2,000 paired learner-cluster bootstrap samples with seed 20261003. Resampling retains all enrollments for each sampled learner and uses the enrollment count as denominator. The fitted models stay fixed, so the intervals reflect evaluation-sample uncertainty conditional on the development models.

Plan and independent checks

The day-56 window was chosen from schedules and cohort availability before activity-mark associations were examined. Version 0.2 added an exact-row-uniqueness sensitivity after the source integrity audit found repetitions. The primary rows and comparison were unchanged. This plan is locally dated, not externally preregistered.

Independent NumPy least-squares reconstruction reproduced all 32 reported MAEs for the main models, sensitivities and additional offering check within 0.00000001 points. Read-only review checked predictor restrictions and bootstrap construction. Source code, fingerprints and execution records are retained.

Small additional predictive value

Model	Held-out MAE	Reduction	95% interval
Prior mark + course offering	11.621	-	-
+ Click volume / primary	11.302	0.319	[0.207, 0.426]
+ Active days / secondary	11.256	0.365	[0.250, 0.470]
+ Both / secondary	11.231	0.390	[0.268, 0.505]

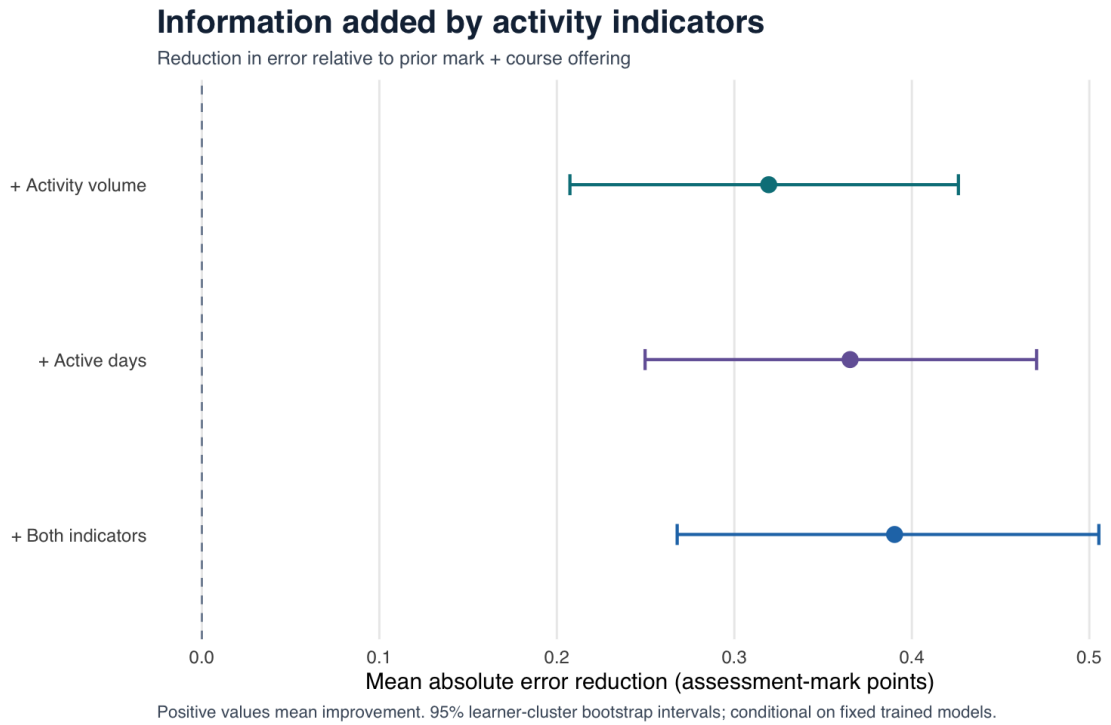


Figure 1. Paired reductions in assessment-mark error relative to the reference model. Intervals are conditional learner-cluster bootstrap intervals; the point estimates are enrollment weighted.

The primary reduction is about 2.75% of the reference MAE. Click volume adds information in this specific setting, while the remaining average error is about 11.30 mark points. An improvement in prediction should not be read as a corresponding increase in a learner's score.

Active-day and combined models have numerically lower error than the reference. Their ordering is secondary; no prespecified direct superiority test between indicators was performed. The study does not select a new primary indicator after seeing this ordering.

The result supports a cautious measurement claim: activity can provide context about later assessed performance. It does not justify interpreting an interaction count as understanding, effort, goal displacement, or mastery.

Context and selection matter

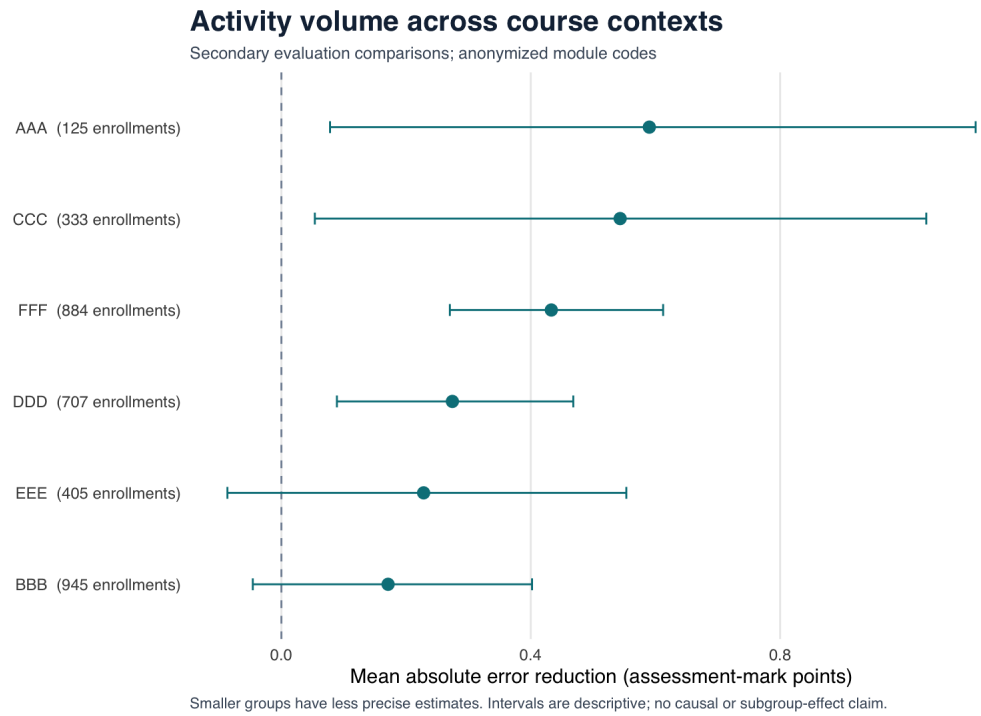


Figure 2. Secondary volume comparisons by anonymized module. All point estimates are positive; the BBB and EEE intervals include no improvement. These are descriptive intervals, not a test of differences between modules or causal heterogeneity.

Course context affects what an interaction represents. Different resources, schedules and assessment designs limit generalization; this concern is well established in learning analytics. [2-4] The primary split reserves learners in offerings represented during fitting, rather than estimating performance in an unseen course offering.

Missing marks differ from observed marks

Reserved evaluation group	Records	Median clicks	Median active days
Observed target mark	3,399	81	6
No target submission record	695	25	2
Submission recorded, mark absent	5	85	6

Learners without a target submission have lower median recorded activity and lower mean prior marks (68.13 versus 76.98 among those with an observed target mark). These diagnostics show selection differences; they do not correct missing outcomes or establish why submission was absent.

Robustness with visible tradeoffs

Volume comparison	Eval marks	Reduction	95% interval
Positive-weight prior only	3,109	0.177	[0.095, 0.263]
Exclude withdrawal on cutoff day	3,399	0.319	[0.207, 0.426]
Quadratic prior/activity terms	3,399	0.198	[0.111, 0.281]
Exact-row-unique click volume	3,399	0.318	[0.208, 0.424]
28-day window / narrower cohort	2,270	0.477	[0.340, 0.623]
Day-70 cutoff / new selection	3,218	0.073	[0.019, 0.128]

How the populations and benchmarks change

Longer window: activity on days 28-55 retains the primary assessment IDs but requires the selected prior submission before day 28. The eligible cohort narrows to 14,016 enrollments. There are 9,370 development and 2,270 evaluation marks; reference MAE is 11.874 points. The larger improvement cannot be attributed solely to a longer window because the population changes.

Later cutoff: day 70 uses activity on days 56-69 and recomputes prior and target selection. Its 18,470 eligible enrollments include 12,868 development and 3,218 evaluation marks; reference MAE is 9.499 points. The smaller improvement is conditional on a potentially different target assessment and population.

Reference form: adding quadratic prior-score terms lowers reference error to 10.438 points. Activity volume then improves that stronger reference by 0.198 points. Incremental value depends partly on how earlier performance is modeled.

Record encoding: 51,964 exact complete-row repetitions occur beyond their first instance in 684,147 eligible-window source rows. Their origin is unknown. Retaining every official row is primary; collapsing exact repetitions changes the volume improvement very little. This is a robustness encoding, not a claim that the repetitions are errors.

Removing withdrawal on day 56 excludes five already-eligible records, none with an observed target mark, so the mark-error estimates remain identical. None of these checks replaces the primary comparison.

A later-offering stress test

Additional secondary analysis

The exact rule was fixed after inspecting primary estimates: train on 2013B, 2013J and 2014B development enrollments; evaluate reserved learners in 2014J. The broader plan anticipated transferable module effects. Offering-specific coefficients and tuning are excluded.

Training contains 8,911 observed enrollments (8,634 learners); evaluation contains 1,189 (1,155 learners), across six modules with zero learner overlap. This reuses primary reserved learners; it is not a newly independent evaluation sample.

Module-effect model	MAE	Reduction	95% interval
Prior mark + module	16.379	-	-
+ Click volume	15.873	0.506	[0.365, 0.650]
+ Active days	15.835	0.544	[0.411, 0.675]
+ Both indicators	15.788	0.591	[0.450, 0.733]

Prediction errors remain large

Volume adds information here, but remaining error is 15.873 points. The different population and module-effect reference preclude a controlled comparison with primary MAE. The large absolute errors warn against assuming reliable future-offering predictions.

2014J module	Evaluation marks	Reference MAE	Volume MAE
AAA	56	7.576	6.956
BBB	290	35.516	34.287
CCC	200	10.632	10.464
DDD	213	10.416	10.050
EEE	158	8.106	7.957
FFF	272	11.488	11.213

BBB has reference error of 35.516 points and volume error of 34.287, much larger than the other modules. These diagnostics require investigation of offering-specific assessment conditions; they do not establish a cause.

The combined model produces one raw prediction above 100 (100.244); other models stay in range. Offering labels do not establish live grade availability.

From an indicator to a study choice

A working, fictional companion

The dashboard demonstration compares the same invented learner record in an activity-only view and an activity-plus-assessment view. Activity values remain synchronized. An unknown assessment stays unknown; earlier marks are explicitly retrospective because release times are unavailable.

A counts-versus-rates mini lesson then lets a visitor answer, repeat, switch questions or read explanations. Two point rules update from the identical in-memory history. Their difference is inspectable rather than hidden in an overall progress percentage.

Shared practice history	All-correct credit	Distinct-item credit
First correct answer to item 1	+1	+1
Correct repeat of item 1	+1	+0
Incorrect answer to item 3	+0	+0
Read item 3 explanation	+0	+0
Total after those four actions	2	1

All-correct credit counts correct response events. Distinct-item credit counts different items answered correctly. Neither is a validated mastery measure. Both policies are shown together for demonstration; a future randomized study would show only the assigned policy while keeping tasks and feedback consistent.

What the current work cannot establish

OULAD contains no dashboard assignment, displayed-score exposure or within-day choices. The analysis therefore cannot identify metric-seeking motives, deliberate gaming, goal displacement, or a causal activity effect. Offline reading and reflection are unobserved. Historical marks are neither learning gains nor independent transfer measures.

A next-stage experiment would test whether the point rule changes repeat-versus-new-item choices and performance on a separate unfamiliar assessment. It would require participant access, an appropriate reviewed protocol, a specified allocation procedure and outcome definitions, and an adequately planned sample. No such experiment has been run.

The current prototype collects no participant data and uses no sign-in, analytics, identifiers or persistent storage. Desktop and narrow-screen browser walkthroughs and scoring checks were completed; assistive-technology evaluation and participant usability testing remain outstanding.

Evidence, sources and next use

A focused research contribution

Activity indicators and early outcome prediction already have substantial precedents. Related OULAD studies examine click features, domain differences and time cutoff leakage; one recent descriptive study uses total clicks and active days directly. [3-6] This piece contributes a transparent, indicator-specific comparison and a companion design argument. It claims no first-study novelty.

Reproducibility and authorship

The package includes acquisition code, integrity checks, versioned local plans, cohort scripts, explicit model formulas, aggregate results, figures, and independent metric verification. The frozen configuration preserves its original pre-analysis status; separate execution records show that evaluation is now complete. Re-running the same reserved set is reproducibility, not new validation.

AI assistance supported literature research, implementation and drafting. Before using the artifact in an application, the applicant should inspect the code and cited evidence, reproduce the result, explain the statistical choices in their own words, and take responsibility for the report. This is an independent portfolio study, not a peer-reviewed publication.

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